what is a research question?

prof. dr. nicolas meseth

1 from topic to question
2 what makes a good one
3 questions without an answer
4 sharpening a question
5 the reality test

a data set with 606 columns answers nothing by itself, until a question tells you which shelf to light up.

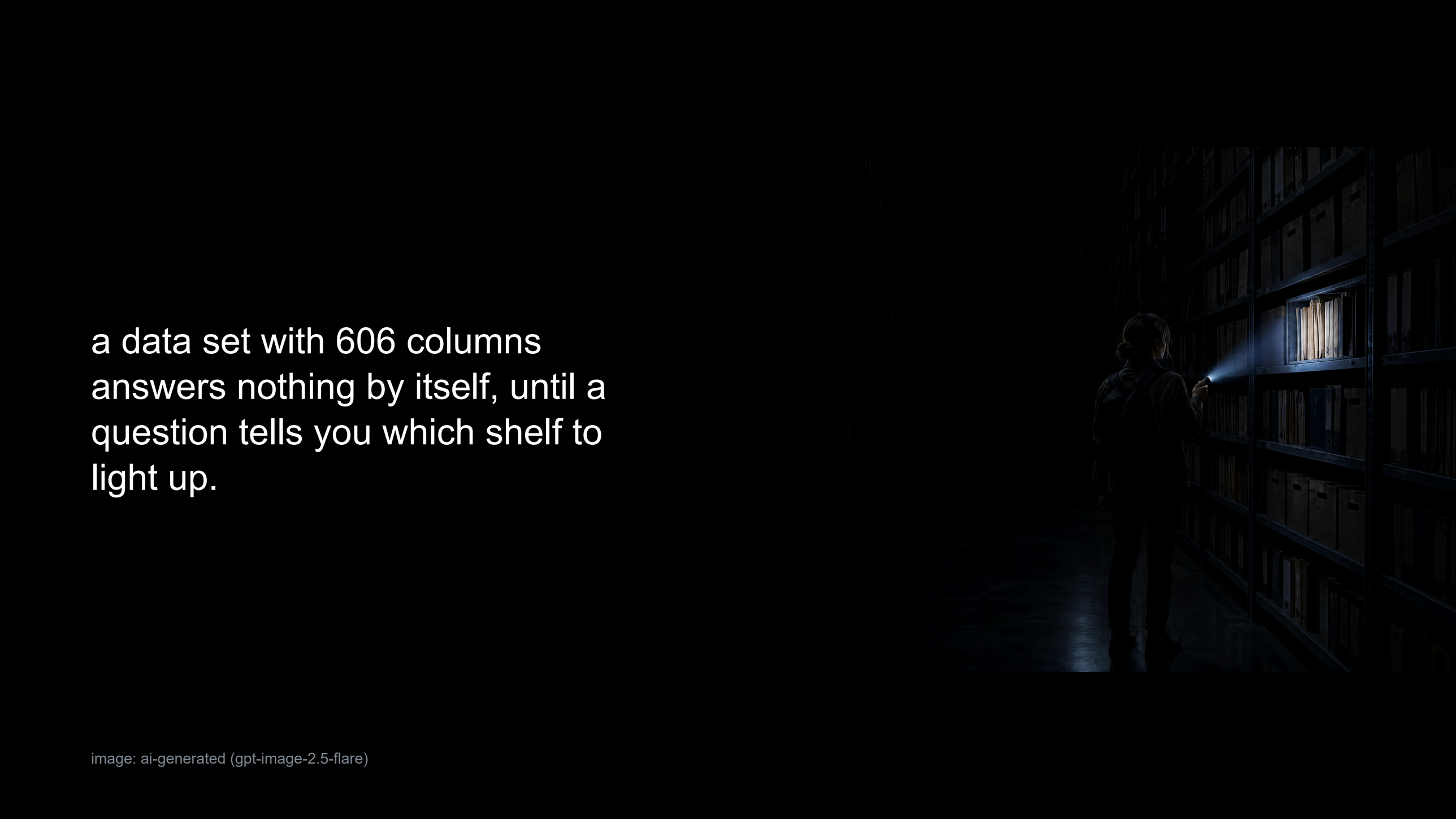


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part 1

from topic to question

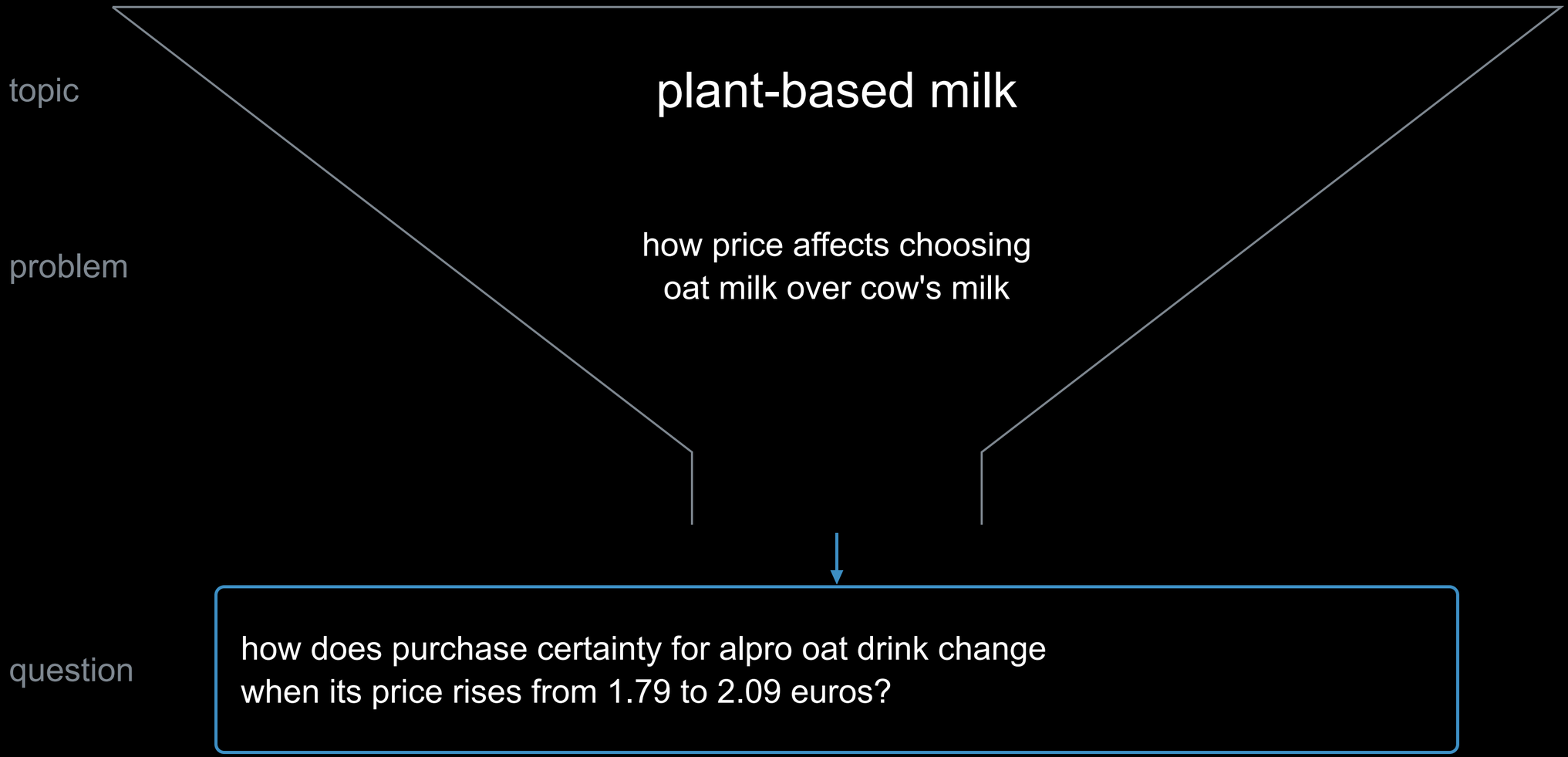
topic



after Döring (2023), Forschungsmethoden und Evaluation, p. 148



after Döring (2023), Forschungsmethoden und Evaluation, p. 148



after Döring (2023), Forschungsmethoden und Evaluation, p. 148

two conditions

answerable with evidence

tells us something about
how the world works

after Huntington-Klein, The Effect, ch. 2

two conditions

answerable with evidence	tells us something about how the world works
which milk ad is the best?	how many respondents live in lower saxony?
no measure for “best”, so no evidence can decide it	answerable, but the answer teaches nothing about behaviour

after Huntington-Klein, The Effect, ch. 2

two conditions

answerable with evidence	tells us something about how the world works
which milk ad is the best?	how many respondents live in lower saxony?
no measure for “best”, so no evidence can decide it	answerable, but the answer teaches nothing about behaviour

do respondents who care about the environment buy a larger share of organic food? ✓

`p030green_5umweltbewusst, v032bioanteil`

after Huntington-Klein, The Effect, ch. 2

which of these is a research question?

sustainability and milk

milk is too cheap

do vegans buy more organic?

analyse the milk survey

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research question

open: it asks and does not claim,
often a w-question (how, which, who),
for when prior knowledge is thin

for example

which shops do vegans
use for their food?

hypothesis

a claim with a direction, derived
from theory or earlier findings,
which you have to cite

for example

respondents over 60 buy a larger
share of organic food than
respondents under 30

with a survey that is already collected, the open question is the normal case

after Döring (2023), Forschungsmethoden und Evaluation, pp. 149–152

part 2

what makes a good one

three kinds of questions

descriptive

how many, how often,
how is it distributed?

relational

do groups differ, do two
things go together?

causal

what happens if we
change one thing?

descriptive

how many, how often, how is it distributed?

what share of the respondents shops at a discounter?

`v008ort_1discount`

relational

do groups differ, do two things go together?

causal

what happens if we change one thing?

descriptive

how many, how often,
how is it distributed?

what share of the
respondents shops
at a discounter?

`v008ort_1discount`

relational

do groups differ, do two
things go together?

do people in big cities
shop online more often?

`d044stadt`,
`v008ort_9online`

causal

what happens if we
change one thing?

descriptive	relational	causal
how many, how often, how is it distributed?	do groups differ, do two things go together?	what happens if we change one thing?
what share of the respondents shops at a discounter?	do people in big cities shop online more often?	does a higher price lower purchase certainty for alpro oat drink? ⚠
`v008ort_1discount`	`d044stadt`, `v008ort_9online`	`ad015p1reaktion4`, `U015p1reaktion4_3alpro`

needs a design that separates the cause

potential results

would every possible answer teach you something?

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feasibility

do the data contain what you need?

a checklist

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feasibility

do the data contain what you need?

scale

can you answer it in the time you have?

after Huntington-Klein, The Effect, 2.4

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design

can these data separate what you want to separate?

after Huntington-Klein, The Effect, 2.4

a checklist

potential results

would every possible answer teach you something?

feasibility

do the data contain what you need?

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can you answer it in the time you have?

design

can these data separate what you want to separate?

simplicity

is it one question, not three?

after Huntington-Klein, The Effect, 2.4

too broad, just right

what drives organic food purchases?

a whole research programme, not one question



one variation

`V041nofleischf`

one outcome

`v032bioanteil`

one comparison

vegetarians against no restrictions

a narrow question names one variation, one outcome and one comparison.

a question you can answer well is worth more than an important one you can only talk about.

part 3

questions without an answer

normative

should oat milk cost as much as cow's milk?

data describe what is, they do not judge it

four traps

normative

should oat milk cost as much as cow's milk?

data describe what is, they do not judge it

vague

what do germans think about food?

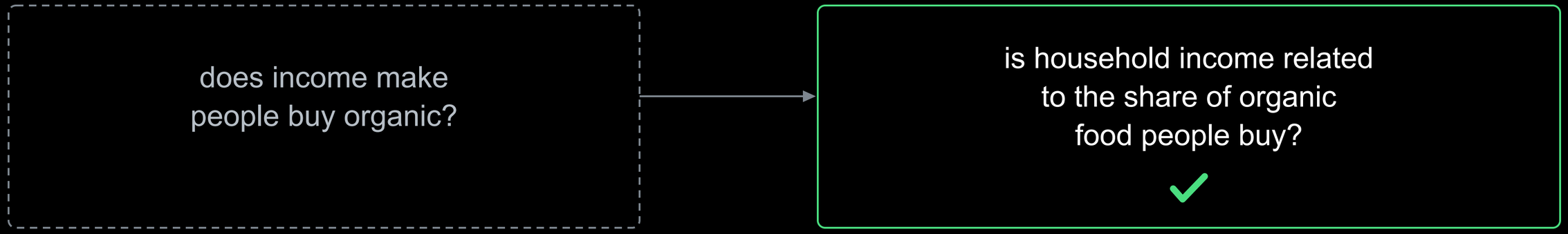
no object, no measure

four traps

normative	should oat milk cost as much as cow's milk?	data describe what is, they do not judge it
vague	what do germans think about food?	no object, no measure
causal without design	does a higher income make people buy organic?	income goes with education, place and values

four traps

normative	should oat milk cost as much as cow's milk?	data describe what is, they do not judge it
vague	what do germans think about food?	no object, no measure
causal without design	does a higher income make people buy organic?	income goes with education, place and values
not in the data	has organic buying grown since 2020?	one survey at one time, november 2025



causal, without a design
that could show it

`d049einkommen, v032bioanteil`

“makes” becomes “is related to”, and the report says honestly why it cannot claim more.

which question can the milk survey answer?

is milk too cheap?

do vegans use farm shops?

did prices rise since 2020?

do ads cause brand loyalty?

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part 4

sharpening a question

from idea to question



from idea to question



from idea to question

prices and oat milk

who	people who bought and drank milk or milk alternatives in the last 12 months
where	an online panel across germany
when	
variation	
outcome	
comparison	

from idea to question

prices and oat milk

who	people who bought and drank milk or milk alternatives in the last 12 months
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from idea to question

prices and oat milk

who	people who bought and drank milk or milk alternatives in the last 12 months
where	an online panel across germany
when	november 2025
variation	price of alpro oat drink: 1.69, 1.79, 1.89 or 2.09 euros
outcome	
comparison	

from idea to question

prices and oat milk

who	people who bought and drank milk or milk alternatives in the last 12 months
where	an online panel across germany
when	november 2025
variation	price of alpro oat drink: 1.69, 1.79, 1.89 or 2.09 euros
outcome	purchase certainty from −4 to +4
comparison	

from idea to question

prices and oat milk

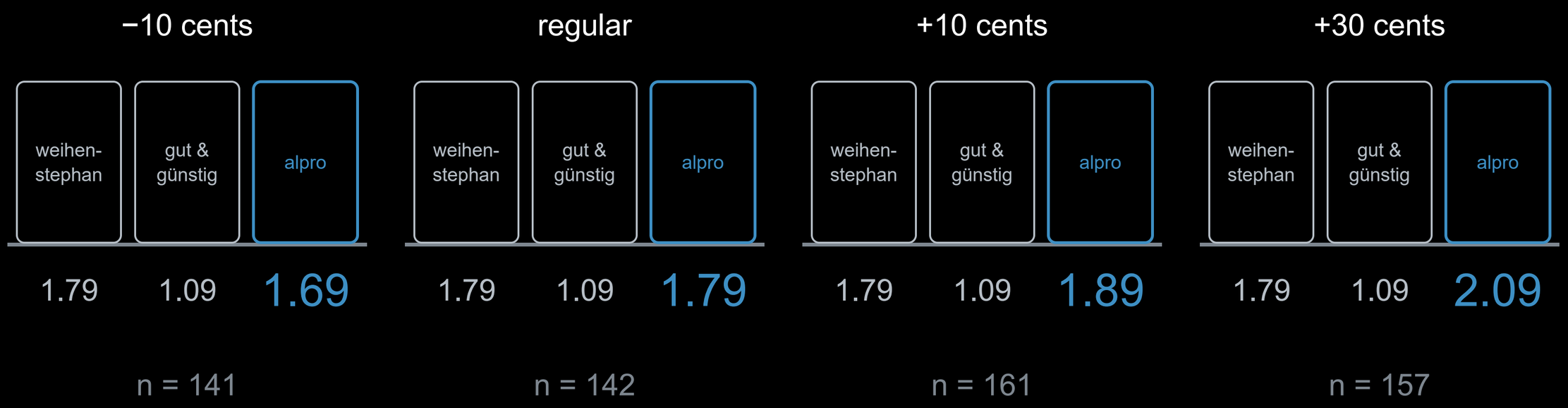
who	people who bought and drank milk or milk alternatives in the last 12 months
where	an online panel across germany
when	november 2025
variation	price of alpro oat drink: 1.69, 1.79, 1.89 or 2.09 euros
outcome	purchase certainty from −4 to +4
comparison	each price against the regular 1.79

the sharpened question

how does purchase certainty for alpro oat drink change among people who buy and drink milk when its price lies 10 cents below, or 10 or 30 cents above, the regular 1.79 euros?

the price experiment

chance decided which shelf each respondent saw



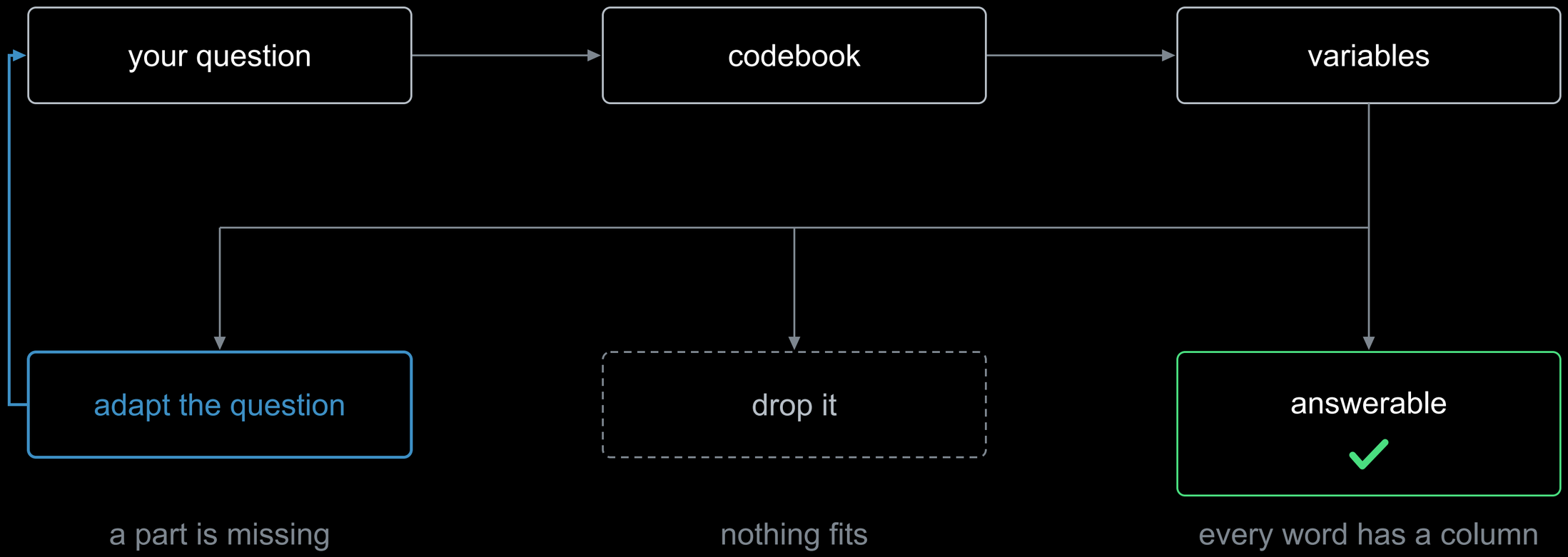
chance made the groups alike except for the price, so a difference in purchase certainty can be put down to it.

part 5

the reality test


your question
codebook
variables

the data came first



a part is missing

nothing fits

every word has a column

check it against the codebook

in the question	column	what the codebook says
people who buy and drink milk	`v006gekauft_3mi == 3`	bought and drank milk or milk alternatives in the last 12 months
price of alpro oat drink	`ad015p1reaktion4`	codes 1 to 4 for 1.69, 1.79, 1.89 and 2.09 euros, assigned by chance
purchase certainty	`U015p1reaktion4_3alpro`	from −4 for certainly not to +4 for certainly
the regular 1.79 euros	`ad015p1reaktion4 == 2`	“`Normal alpro`” in `AD015p1reaktion4f`
online, november 2025	no column	the study documentation

601 respondents saw a price and 596 of them rated alpro; the test went only to people who bought and drank milk, although the questionnaire also names those who only bought it.

write your question down before you compute anything, check every word against the codebook, and change the question rather than the data.



image: ai-generated (gpt-image-2.5-flare)